LIFT-CF: LIFTING COUNTERFACTUAL EXPLANATIONS TO DISTRIBUTIONAL ROBUSTNESS

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ABSTRACT

Counterfactual explanations (CFs) recommend actionable changes an individual can make to flip a machine learning model's prediction toward a desired outcome. Most CF generators assume a static model and data distribution, yet models deployed in practice are retrained periodically, so a CF that is valid today may not be tomorrow. Institutions that deploy these models typically issue a population of CFs, so robustness is fundamentally a population-level property, not an individual-level one. Existing robustness methods, however, certify or optimize validity of each CF individually, leaving this population-level risk unaddressed. We introduce LIFT-CF, a distributionally robust wrapper that treats the population of CFs produced by any CF generator as the object to be made robust, rather than any single CF. LIFT-CF constructs a Wasserstein ambiguity set anchored to both the observed target-class distribution and the generator's own CF output, and shows that minimizing a tractable upper bound on the worst-case risk over this set reduces to moving the CF distribution along a calibrated Wasserstein barycentric path. Extensive experiments across ... show that ...

1 INTRODUCTION

Counterfactual explanations (CFs) help users identify changes that would alter a machine learning model's prediction. During the generation process, most methods assume a fixed model and data distribution; in deployment, however, the setting is considerably more dynamic (Quionero-Candela et al., 2009). Evolving data distributions or policy changes cause retraining, and previously generated CFs might lose their validity in the new setting (Rawal et al., 2020). A counterfactual explanation that recommends actions to alter a model's prediction is only useful if the action still delivers the desired outcome for a model that might have been updated in between. When it does not, the recourse for a user, for example, an applicant told to raise their income by $ 3000 to secure loan approval, expires, undermining the trustworthiness and utility of CFs in dynamic environments (Figure 1).

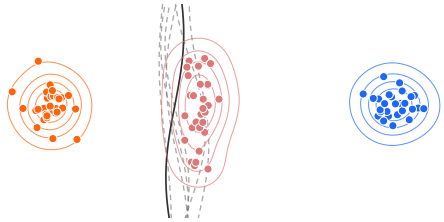


Figure 1: **Illustrative example of decision boundary shift.** A population of counterfactuals (pink) sits near a decision boundary (solid curve) separating two class distributions (orange and blue). Dashed lines indicate how the boundary's position could shift, illustrating why a counterfactual valid under the current boundary may become invalid under a shifted one.

This concern is not only at an individual level but also distributional. A bank, for instance, does not provide a single counterfactual but a population of them, one per rejected applicant. When the underlying data distribution shifts and the model is retrained, it is not only one CF but a fraction of the previously generated population that becomes invalid. From an institutional perspective, the relevant object of robustness is not the individual CF but the distributional risk of invalidity across the entire generated population, a risk that instance-by-instance certification cannot control, since it never represents the population as an object of its own.

Current strategies aim to ensure that a counterfactual remains valid despite model updates. One line of work constrains these changes *implicitly*, without directly modeling how the model might

change: counterfactuals are optimized for high classification certainty (Dutta et al., 2022; Hamman et al., 2023; Jiang et al., 2023b), training data is augmented with CFs (Ferrario & Loi, 2022), or robustness is certified a posteriori under a Bayesian view over a user-specified pool of retrained models (Stepka et al., 2025; Duell & Fan, 2026). Another line of work models the changes *explicitly*, bounding against plausible parameter shifts for a single CF Upadhyay et al. (2021); Kayastha et al. (2024); Kyaw et al. (2025); Marzari et al. (2024); Jiang et al. (2023a), including against a mixture of plausible future parameter distributions (Nguyen et al., 2023). Two works come closer to a population view: Bui et al. (2022) jointly optimize a small per-instance set of CFs for joint validity under a shifted model, while Nguyen et al. (2022) perturb the data distribution directly but still certify recourse one input at a time. Neither treats the population generated across instances as the object of robustness.

We address this gap with LIFT-CF, a post-hoc distributional refinement that treats the population of CFs produced by any CF generator as a distribution to be moved, via a calibrated Wasserstein lens anchored to the target class and the generator's output, toward a region of lower worst-case future invalidity. Our main contributions are:

- We introduce the *CF-induced Wasserstein lens*, an ambiguity set of target-side distributions close to the observed target class and generator output, giving a worst-case notion of distributional lens risk that a refined CF population can be optimized against.
- We show that minimizing a tractable convex relaxation of this worst-case risk reduces to a one-dimensional Wasserstein barycentric family between the CF and target distributions, and select the deployed refinement by directly evaluating robustness to adversarial retraining shifts on held-out data, with a matching Lipschitz stability guarantee bounding the resulting finite-grid search error.
- We realize the calibrated refinement via optimal transport and show empirically that it experiments?

2 Related Work

Model parameter updates typically arise from retraining on new data Rawal et al. (2020). Existing approaches constrain such changes either *implicitly*, without modeling their source, or *explicitly*, by model plausible parameter shifts .

Implicit constraints. The methods optimize CFs for high classification certainty Dutta et al. (2022); Hamman et al. (2023); Jiang et al. (2023b), augment training data with generated CFs Ferrario & Loi (2022), or perturb the covariance of data sampled near the decision boundary to train a linear surrogate Bui et al. (2025). Xu & Wu (2024) relax the naturally-occurring model change condition of Hamman et al. (2023) to a subgaussian bound. Two works certify robustness a posteriori under a Bayesian view over a user-specified pool of retrained models: Stepka et al. (2025) move a base CF into a region meeting a specified robustness level, while Duell & Fan (2026) optimize on the generation side for high confidence and low variance in the CF's posterior predictive distribution, certifying graceful degradation via a KL-divergence bound.

Explicit constraints. Upadhyay et al. (2021) introduce ROAR, a minimax objective generating CFs that minimize the worst-case plausible shift under a linear surrogate model; Kayastha et al. (2024) give a provably optimal solver for any generalized linear model, and Kyaw et al. (2025) extend this to general L_p-bounded neighborhoods. Nguyen et al. (2023) similarly hedge against plausible parameter shifts via a min-max problem, where the max term is the Gelbrich distance to an ambiguity set of future model-parameter distributions, solved by gradient descent. Bui et al. (2022) instead bound the joint validity probability of a set of CFs. APΔS Marzari et al. (2024; 2025) samples plausible model shifts directly and shows exact robustness decisions are NP-hard. Jiang et al. (2023a; 2024b) propagate interval-valued weights and biases through the network to bound hidden-unit activations under bounded parameter perturbations. These parameter-space approaches assume model changes stem from data shifts without modeling the shift itself. Nguyen et al. (2022) address this by adversarially perturbing the data via a Gaussian-mixture ambiguity set over Wasserstein distance on the mean-covariance space. Two works come closest to a population view: Bui et al. (2022) jointly optimize a small per-instance CF set for joint validity, while Nguyen et al. (2022), despite modeling

the data shift, still certifies recourse one input at a time. Neither treats the population of generated CFs as the robustness object, the gap our method closes.

A related line of work study robustness to *model multiplicity* Breiman (2001); Marx et al. (2020); Black et al. (2022); Rudin (2019); Coston et al. (2021); Pawelczyk et al. (2020); Turbal et al. (2026), where distinct near-optimal models yield inconsistent CFs, with no assumption on how or why models differ. See Jiang et al. (2024a) for a broader robustness survey.

3 Distributionally Robust Counterfactuals

We propose LIFT-CF, a distributional robustness wrapper for CF generators that operates *post-hoc* on the population of CFs a generator produces, rather than modifying its objective or parameters. We first state the exact robustness objective and show it is intractable to optimize directly (Section 3.1). We then construct a CF-induced Wasserstein lens, a tractable, data-level relaxation that captures where the CF distribution should lie to remain valid under future shifts, and show under an explicit regularity condition that it provably controls the exact objective (Section 3.2). The robust relaxation of the resulting minimax objective induces a one-dimensional Wasserstein barycentric family (Section 3.3), on which we select the deployed operating point by evaluating the true objective directly, using held-out proxy deployment shifts (Section 3.4). We show this one-dimensional objective is Lipschitz along the path and bound the resulting finite-grid search error (Section 3.5). Finally, we realize the selected refinement via optimal transport (Section 3.6).

3.1 Problem Setting and the Robust Objective

Let $f_0 : \mathcal{X} \to [0,1]$ be a binary classifier and $\tau \in (0,1)$ a decision threshold; x is *favorable* if $f_0(x) \geq \tau$ and *unfavorable* otherwise. Write $\ell_f(z) = [\tau - f(z)]_+$ for the margin-based counterfactual invalidity loss. Let G be a base CF generator. Given a batch $X = \{x_i\}_{i=1}^n$ of instances with $f_0(x_i) < \tau$, let $z_i = G(x_i)$ denote the corresponding CFs for a favorable outcome. We represent the generated CF population by its empirical distribution $\mathbb{P}_{\mathrm{CF}} = \frac{1}{n}\sum_{i=1}^n \delta_{z_i}$, and let $\mathbb{P}_1$ denote the empirical distribution of target-class training samples ($f_0(x) \geq \tau$). The two distributions provide complementary information: $\mathbb{P}_{\mathrm{CF}}$ captures where the base generator currently places CFs, while $\mathbb{P}_1$ captures the observed structure of the favorable class. We use consistency to denote closeness to $\mathbb{P}_{\mathrm{CF}}$, and *plausibility* to denote alignment with $\mathbb{P}_1$.

A CF favorable under f_0 does not necessarily remain favorable once f_0 is updated: a shift in the decision boundary may move the CF back into the unfavorable region. Since the future classifier is unobserved at deployment time, we cannot hedge against a single predicted shift. For a Wasserstein uncertainty radius $r \geq 0$, define the data-side uncertainty set $\mathcal{U}_r = \{\mathbb{P} : W_{\mathrm{data}}(\mathbb{P}, \widehat{\mathbb{P}}_{\mathrm{tr}}) \leq r\}$, and for $\mathbb{P} \in \mathcal{U}_r$, let $f_{\mathbb{P}} = \mathrm{Update}(f_0, \mathbb{P})$ denote the classifier obtained by applying the fixed model-update procedure (fine-tuning from f_0, in our implementation) to the shifted training distribution. The exact robust objective for a candidate refined distribution $\mathbb{Q}$ is

$$\Phi_r(\mathbb{Q}) := \sup_{\mathbb{P} \in \mathcal{U}_r} \mathbb{E}_{z \sim Q}\left[\ell_{f_{\mathbb{P}}}(z)\right], \tag{1}$$

and the ideal refinement solves $\min_{\mathbb{Q} \in \mathcal{P}_2(\mathcal{X})} \Phi_r(\mathbb{Q})$.

This objective is generally intractable: the inner supremum requires a full model update for every candidate, and the outer minimization ranges over $\mathcal{P}_2(\mathcal{X})$, with no closed form linking a candidate $\mathbb{Q}$ to the retrained scores it will face. We instead build a tractable data-level relaxation, the CF-induced Wasserstein lens, and show it provably controls Φ_r (Sections 3.2). This leads to a population-level rather than instance-level view of robustness: instead of independently robustifying each CF, we seek a refined *distribution* whose lens risk provably bounds Φ_r, leaving G itself unchanged.

3.2 CF-Induced Wasserstein Lens

We model future distributions as bounded W_2-displacements of the distributions observable today. For a reference distribution $\mathbb{P}$ and radius $r \geq 0$, the Wasserstein ball $B_{W_2}(\mathbb{P}, r) = \{\mu \in \mathcal{P}_2(\mathcal{X}) : W_2(\mu, \mathbb{P}) \leq r\}$, contains distributions reachable from $\mathbb{P}$ at transport cost at most r, which we regard as a plausible future distribution around $\mathbb{P}$.

Definition 1 (CF-induced Wasserstein lens)**.** Let $\mathbb{P}_1$ be the empirical class-conditional distribution and $\mathbb{P}_{\mathrm{CF}}$ the CF distribution, moving instances from unfavorable to favorable outcomes. We define the *CF-induced Wasserstein lens*

$$\mathcal{C} = \{\mu \in \mathcal{P}_2(\mathcal{X}) : W_2(\mu, \mathbb{P}_1) \le \epsilon,\ W_2(\mu, \mathbb{P}_{\mathrm{CF}}) \le \delta\}\,. \tag{2}$$

$\mathcal{C}$ is the set of plausible future valid CF distributions, anchored jointly on the consistent ($\mathbb{P}_{\mathrm{CF}}$) and the plausible evidence ($\mathbb{P}_1$); see Figure 2 for a schematic construction. Neither anchor alone suffices: $\mathbb{P}_1$ keeps the refinements close to the favorable class samples, but ignores the generator and its current CFs, and $\mathbb{P}_{\mathrm{CF}}$ keeps the refinement close to what the generator already produced, but does not include signal about proximity to class 1 samples. The two are also not independent: shifts that alter the class population can simultaneously induce changes in the learned decision boundary, and hence in the location and validity geometry of the generated CFs, which is why we impose both signals jointly.

Proposition 1 (Wasserstein consistency–plausibility trade-off)**.** *Let $D = W_2(\mathbb{P}_{\mathrm{CF}}, \mathbb{P}_1)$ and let $\{\mathbb{P}_t^{\mathrm{geo}}\}_{t\in[0,1]}$ be any constant-speed W_2-geodesic connecting $\mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1$, with $\mathbb{P}_0^{\mathrm{geo}} = \mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1^{\mathrm{geo}} = \mathbb{P}_1$. Then, $W_2(\mathbb{P}_t^{\mathrm{geo}}, \mathbb{P}_{\mathrm{CF}}) = tD$ and $W_2(\mathbb{P}_t^{\mathrm{geo}}, \mathbb{P}_1) = (1-t)D$ for every $t \in [0,1]$, so $\mathbb{P}_t^{\mathrm{geo}} \in \mathcal{C}$ iff $tD \le \delta$ and $(1-t)D \le \epsilon$, i.e.,*

$$t \in \left[1 - \frac{\epsilon}{D}, \frac{\delta}{D}\right] \cap [0,1].$$

This interval, and hence the lens's intersection with the geodesic, is nonempty iff $\epsilon + \delta \ge D$, collapsing to the unique parameter $t^* = \frac{\delta}{D} = 1 - \frac{\epsilon}{D}$, when equality holds.

Proposition 1 gives the lens a direct geometric interpretation: radii δ and ϵ are not two independent notions of consistency and plausibility, but the two radii must together cover the geodesic distance D between the anchors for $\mathcal{C}$ to be nonempty, an assumption we make throughout ($\epsilon + \delta \ge D$).

Role of ϵ and δ. δ bounds how far a plausible future CF distribution may drift from what the generator currently produces, and ϵ bounds how far it may sit from the currently observed favorable class.

Robustness. Robustness is measured by requiring the refined CF distribution $\mathbb{Q}$ to stay close to every plausible future CF distribution μ in the lens. For any candidate $\mathbb{Q} \in P_2(\mathcal{X})$, define its worst-case lens risk as

$$R(\mathbb{Q}) = \sup_{\mu\in\mathcal{C}} W_2^2(\mathbb{Q}, \mu). \tag{3}$$

A smaller $R(\mathbb{Q})$ indicates a smaller worst-case distributional mismatch over the uncertainty lens.

We now connect this data-level quantity to Φ_r under an explicit regularity condition. Let $\mathcal{F}_r^{\mathrm{retr}} := \{f_{\mathbb{P}} = \mathrm{Update}(f_0, \mathbb{P}) : \mathbb{P} \in \mathcal{U}_r\}$.

Assumption 1 (Stability of Retraining). Every $f_{\mathbb{P}} \in \mathcal{F}_r^{\mathrm{retr}}$ is L_r-Lipschitz uniformly over $\mathbb{P} \in \mathcal{U}_r$: $L_r = \sup_{\mathbb{P}\in\mathcal{U}_r} \mathrm{Lip}(\ell_{f_P}) < \infty$. Equivalently, $\ell_{f_{\mathbb{P}}}$ is L_r-Lipschitz uniformly over $\mathcal{U}_r$.

This is a concrete condition on Update.

Lemma 1 (Uniform score-difference bound)**.** *Under Assumption 1, for any $\mu, \nu \in \mathcal{P}_2(\mathcal{X})$ and any $\mathbb{P} \in \mathcal{U}_r$, $|\mathbb{E}_\mu[\ell_{f_{\mathbb{P}}}] - \mathbb{E}_\nu[\ell_{f_{\mathbb{P}}}]| \le L_r\, W_2(\mu, \nu)$.*

Proof. Kantorovich–Rubinstein and the L_r-Lipschitzness of $\ell_{f_{\mathbb{P}}}$ give $|\mathbb{E}_\mu[\ell_{f_{\mathbb{P}}}] - \mathbb{E}_\nu[\ell_{f_{\mathbb{P}}}]| \le L_r W_1(\mu, \nu)$; the claim follows from $W_1 \le W_2$. □

We reuse Lemma 1 below to connect $R(\mathbb{Q})$ to Φ_r, and again in Section 3.5 to bound Φ_r's sensitivity to λ, both are instances of the same bound applied to different pairs of distributions.

Assumption 2 (Future-valid reference). For every $\mathbb{P} \in \mathcal{U}_r$ there exists $\mu_{\mathbb{P}} \in \mathcal{C}$ and $\alpha_{\mathbb{P}} \ge 0$ with $\mathbb{E}_{z\sim\mu_{\mathbb{P}}}[\ell_{f_{\mathbb{P}}}(z)] \le \alpha_{\mathbb{P}}$. Set $\bar{\alpha}(r) = \sup_{\mathbb{P}\in\mathcal{U}_r} \in [0, \tau]$.

Proposition 2 (Lens risk bounds future invalidity)**.** *Under Assumptions 1 and 2, for any $\mathbb{Q} \in \mathcal{P}_2(\mathcal{X})$,*

$$\Phi_r(\mathbb{Q}) \ \le\ \bar{\alpha}(r) + L_r\sqrt{R(\mathbb{Q})}. \tag{4}$$

Here $\bar{\alpha}(r)$ reflects whether the lens contains low-invalidity references for the retrained scores it must withstand — a property fixed by the lens's own content, not by the choice of $\mathbb{Q}$.

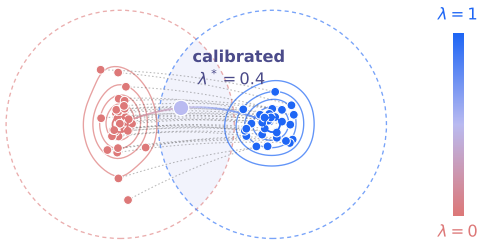


Figure 2: **Certified lens and optimal-transport coupling.** The dashed ellipses show the certified lens $\mathcal{C}$, centered at base CF-distribution $\mathbb{P}_{\mathrm{CF}}$ (pink) and observed target-class distribution $\mathbb{P}_1$ (teal) with radii δ and ϵ. The shaded intersection is $\mathcal{C}$. Dashed lines show the optimal coupling $\pi^\star$ between the two empirical distributions, and the purple marker shows the calibrated barycenter $\mathbb{Q}(\lambda^\star)$ realized along this coupling, positioned inside the lens.

Proof Sketch. Fix $\mathbb{P} \in \mathcal{U}_r$ with reference $\mu_{\mathbb{P}}$ from Assumption 2. Lemma 1 gives $\mathbb{E}_{\mathbb{Q}}[\ell_{f_{\mathbb{P}}}] \leq \mathbb{E}_{\mu_{\mathbb{P}}}[\ell_{f_{\mathbb{P}}}] + L_r W_2(\mathbb{Q}, \mu_{\mathbb{P}}) \leq \alpha_{\mathbb{P}} + L_r\sqrt{R(\mathbb{Q})}$, using $\mu_{\mathbb{P}} \in \mathcal{C}$ and the definition of R. Take the supremum over $\mathbb{P} \in \mathcal{U}_r$. □

Eq. 4 motivates restricting the search to minimizers of $R(\mathbb{Q})$: any refinement that reduces $R(\mathbb{Q})$ tightens the guarantee on Φ_r, up to the fixed residual $\bar{\alpha}(r)$.

3.3 Robust-Bound-Induced Barycentric Geometry

Directly solving

$$\min_{\mathbb{Q}\in\mathcal{P}_2(\mathcal{X})} R(\mathbb{Q}) = \min_{\mathbb{Q}\in\mathcal{P}_2(\mathcal{X})} \sup_{\mu\in\mathcal{C}} W_2^2(\mathbb{Q}, \mu) \tag{5}$$

is generally intractable, since it requires solving an inner supremum over $\mathcal{C}$ for every candidate $\mathbb{Q}$. We instead minimize a tractable family of valid upper bounds on $R(\mathbb{Q})$, which induce a two-anchor Wasserstein barycentric family.

Proposition 3 (Robust-bound-induced barycentric family)**.** *Given the lens $\mathcal{C}$ in Eq. 2, there exists a family of valid upper bounds on the worst-case risk $R(\mathbb{Q})$ whose minimizers take the form*

$$(1-\lambda)W_2^2(\mathbb{Q}, \mathbb{P}_{\mathrm{CF}}) + \lambda W_2^2(\mathbb{Q}, \mathbb{P}_1), \qquad \lambda \in [0, 1]. \tag{6}$$

The minimizers of these relaxed bounds form the two-anchor Wasserstein family

$$\mathcal{Q} := \{\mathbb{Q}(\lambda) : \lambda \in [0, 1]\},$$

where

$$\mathbb{Q}(\lambda) \in \arg\min_{\mathbb{Q}\in\mathcal{P}_2(\mathcal{X})} \left\{(1-\lambda)W_2^2(\mathbb{Q}, \mathbb{P}_{\mathrm{CF}}) + \lambda W_2^2(\mathbb{Q}, \mathbb{P}_1)\right\}, \tag{7}$$

with endpoints $\mathbb{Q}(0) = \mathbb{P}_{\mathrm{CF}}$ and $\mathbb{Q}(1) = \mathbb{P}_1$.

Proof sketch. For any $\mu \in \mathcal{C}$, the lens constraints and triangle inequality give $W_2(\mathbb{Q}, \mu) \leq \min\{W_2(\mathbb{Q}, \mathbb{P}_{\mathrm{CF}}) + \delta,\, W_2(\mathbb{Q}, \mathbb{P}_1) + \epsilon\}$. A convex relaxation followed by Young's inequality yields a family of valid upper bounds whose $\mathbb{Q}$-dependent terms are positive weighted sums of $W_2^2(\mathbb{Q}, \mathbb{P}_{\mathrm{CF}})$ and $W_2^2(\mathbb{Q}, \mathbb{P}_1)$. After normalization, their minimizers take the barycentric form in Eq. 6. The admissible bounds span $\lambda \in [0, 1]$. See Appendix A.1 for the full proof. □

Each $\mathbb{Q}(\lambda)$ minimizes a valid upper bound on $R(\mathbb{Q})$, not necessarily $R(\mathbb{Q})$ itself. The remaining question is therefore not where to search for a robust refinement, but where to locate the refinement along this certified path, which we answer by evaluating Φ_r directly, rather than continuing to rely on the bound.

Corollary 1 (Geodesic characterization). Let $D = W_2(\mathbb{P}_{\mathrm{CF}}, \mathbb{P}_1)$. At the endpoints, $\mathbb{Q}(0) = \mathbb{P}_{\mathrm{CF}}$ and $\mathbb{Q}(1) = \mathbb{P}_1$ trivially. For any $\lambda \in (0, 1)$, every minimizer of Eq. 7 lies at time λ on a constant-speed W_2 geodesic between $\mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1$. In particular,

$$W_2(\mathbb{Q}(\lambda), \mathbb{P}_{\mathrm{CF}}) = \lambda D, \qquad W_2(\mathbb{Q}(\lambda), \mathbb{P}_1) = (1-\lambda)D. \tag{8}$$

If the optimal coupling between $\mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1$ is unique, then $\mathbb{Q}(\lambda)$ is unique for every $\lambda \in (0,1)$, and the barycentric parameter λ coincides with the consistency–plausibility parameter of Proposition 1.

Combining Corollary 1 with Proposition 1 gives $\mathbb{Q}(\lambda) \in \mathcal{C} \iff \lambda D \le \delta$ and $(1-\lambda)D \le \epsilon$.

Geometry versus certificate. The two anchors $\mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1$ determine the *direction* of the barycentric refinement: $\lambda = 0$ recovers the original CF distribution, $\lambda = 1$ gives full alignment with target class distribution, and intermediate values trade off generator-specific CF geometry against target-side alignment. The lens radii ϵ and δ instead provide the theoretical certificate: they determine the admissible consistency–plausibility range and the resulting robust-risk guarantees, without affecting which geodesic the refinement moves along or requiring any numerical instantiation. The decoupling matters in operation: it means that the search direction is fixed by the generator and the data as soon as G and $\mathbb{P}_1$ are given, while the choice of *where* along that direction to certify is a separate, data-driven calibration problem.

3.4 Data-Side Wasserstein Robust Operating-Point Selection

We select the deployed λ by evaluating Φ_r directly on the reduced family $\{\mathbb{Q}(\lambda) : \lambda \in [0,1]\}$, writing $\Phi_r(\lambda) := \Phi_r(\mathbb{Q}(\lambda))$, rather than continuing to rely on the proxy R. Consequently, we do not need a separate R-based suboptimality certificate here: Proposition 2 already certifies that reducing R tightens the bound on Φ_r, and Section 3.5 bounds the grid-search error directly against Φ_r, the quantity actually optimized. For a fixed radius r and refinement depthλ, the robust operating point is

$$\lambda_r^\star \in \arg\min_{\lambda\in[0,1]} \Phi_r(\lambda), \qquad \Phi_r(\lambda) = \sup_{\mathbb{P}\in\mathcal{U}_r} \mathbb{E}_{z\sim\mathbb{Q}(\lambda)}\left[\ell_{f_{\mathbb{P}}}(z)\right].$$

Approximate inner maximization. The supremum defining $\Phi_r(\lambda)$ is generally intractable because each candidate distribution $\mathbb{P}$ induces a model update before its counterfactual loss can be evaluated. We therefore approximate the inner problem by constructing, for every pair (r,λ), an adversarial training distribution $\mathbb{P}_{r,\lambda}^{\mathrm{adv}} \approx \arg\max_{\mathbb{P}\in\mathcal{U}_r} \mathbb{E}_{z\sim Q(\lambda)}\left[\ell_{\mathrm{Update}(f_0,\mathbb{P})}(z)\right]$. $f_{r,\lambda}^{\mathrm{adv}} = \mathrm{Update}\left(f_0, \mathbb{P}_{r,\lambda}^{\mathrm{adv}}\right)$. Importantly, the adversarial distribution is conditioned on both the uncertainty radius and the refinement depth. Different members $Q(\lambda)$ of the barycentric family may therefore induce different worst-case future shifts.

Held-out estimation and cross-radius selection. On a held-out calibration split $\mathcal{D}_{\mathrm{cal}}$, not used to construct the adversarial training distributions, we estimate $\widehat{\Phi}_r(\lambda) = \frac{1}{|I_{\mathrm{cal}}^{\mathrm{valid}}|}\sum_{i\in I_{\mathrm{cal}}^{\mathrm{valid}}}\left[\tau - f_{r,\lambda}^{\mathrm{adv}}\left(z_i^{\mathrm{cal}}(\lambda)\right)\right]_+$ over a grid $\Lambda \subset [0,1]$ and take $\widehat{\lambda}_r \in \arg\min_{\lambda\in\Lambda}\widehat{\Phi}_r(\lambda)$, as the empirical robust operating point. The resulting configurations $\{(r,\widehat{\lambda}_r) : r \in \mathcal{R}\}$ are subsequently evaluated on a disjoint shift-validation split to select the final uncertainty radius $r^\star$. The deployed refinement depth is then fixed as $\lambda^\star = \widehat{\lambda}_{r^\star}$.

3.5 Stability of the One-Dimensional Robust Objective

Section 3.4 approximates $\lambda_r^\star = \arg\min_{\lambda\in[0,1]}\Phi_r(\lambda)$ by evaluating Φ_r on a finite grid $\Lambda \subset [0,1]$ rather than solving the continuous minimization exactly. This only holds if Φ_r varies with λ in a controlled manner; we thus now show that Φ_r is Lipschitz continuous in λ, and derive an explicit bound on the supoptimality introduced by the finite-grid search.

Lipschitz continuity of Φ_r in λ. Recall L_r from Assumption 1 (the uniform sensitivity of any reachable future score to displacement of its input) and $D = W_2(\mathbb{P}_{\mathrm{CF}}, \mathbb{P}_1)$ from Proposition 1 (the length of the barycentric path).

Proposition 4 (Lipschitz stability of the robust objective)**.** *For any $\lambda, \lambda' \in [0,1]$, $|\Phi_r(\lambda) - \Phi_r(\lambda')| \le L_r D|\lambda - \lambda'|$. Hence, Φ_r is Lipschitz continuous on $[0,1]$ and therefore admits at least one robust operating point $\lambda_r^\star \in \arg\min_{\lambda\in[0,1]}\Phi_r(\lambda)$.*

Proof sketch. Fix any $\mathbb{P} \in \mathcal{U}_r$. Lemma 1 bounds the resulting score difference between two points on the path: $|\mathbb{E}_{Q(\lambda)}[\ell_{f_{\mathbb{P}}}] - \mathbb{E}_{Q(\lambda')}[\ell_{f_{\mathbb{P}}}]| \le L_r\, W_2(Q(\lambda), Q(\lambda'))$. Because $\{Q(\lambda)\}$ is a constant-speed

geodesic (Corollary 1), $W_2(Q(\lambda), Q(\lambda')) = D|\lambda - \lambda'|$ exactly, so the bound becomes $L_r D|\lambda - \lambda'|$. This holds for every $\mathbb{P} \in \mathcal{U}_r$, so it also holds after taking the supremum over $\mathbb{P}$ that defines Φ_r. □

Suboptimality of the finite-grid search. In practice, the operating point is selected over a finite grid $\Lambda_h \subset [0,1]$, of maximum spacing h, using the approximate robust risk $\widehat{\Phi}_r(\lambda)$ obtained from the inner adversarial solver. Assume that its uniform estimation error over the search grid satisfies $\max_{\lambda \in \Lambda_h} \left|\widehat{\Phi}_r(\lambda) - \Phi_r(\lambda)\right| \le \eta$. Let $\widehat{\lambda}_r \in \arg\min_{\lambda \in \Lambda_h} \widehat{\Phi}_r(\lambda)$.

Proposition 5 (Search error on the reduced decision path)**.** *Under the assumptions above,* $\Phi_r(\widehat{\lambda}_r) \le \min_{\lambda \in [0,1]} \Phi_r(\lambda) + 2\eta + \frac{L_r D h}{2}$.

Proof sketch. Let $\lambda_r^\star$ be a minimizer of Φ_r on $[0,1]$, and let $\widetilde{\lambda} \in \Lambda_h$ be the closest grid point to $\lambda_r^\star$. Then $|\widetilde{\lambda} - \lambda_r^\star| \le \frac{h}{2}$. By Lipschitz stability, $\Phi_r(\widetilde{\lambda}) \le \Phi_r(\lambda_r^\star) + \frac{L_r D h}{2}$. Using the uniform approximation error and the optimality of $\widehat{\lambda}_r$ on the grid,

$$\Phi_r(\widehat{\lambda}_r) \le \widehat{\Phi}_r(\widehat{\lambda}_r) + \eta \le \widehat{\Phi}_r(\widetilde{\lambda}) + \eta \le \Phi_r(\widetilde{\lambda}) + 2\eta,$$

which gives the stated bound. □

Interpretation. The barycentric geometry derived in Section 3.2 does not itself define deployment robustness; instead, it provides a one-dimensional decision path on which the actual data-side Wasserstein robust risk is stable and searchable. The bound above separates the two sources of numerical suboptimality:

$$\underbrace{2\eta}_{\text{inner-solver approximation}} + \underbrace{\frac{L_r D h}{2}}_{\text{one-dimensional discretization}}.$$

Thus, finer search grids and stronger inner adversarial optimization systematically tighten the quality of the selected operating point.

3.6 OT Barycentric Realization

Given the calibrated operating point $\lambda^\star$, we realize $Q(\lambda^\star)$ via optimal transport. Let $\{z_i\}_{i=1}^n$ and $\{y_j\}_{j=1}^m$ denote the empirical supports of $\mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1$, and $\mathcal{F}_0 = \{x : f_0(x) \ge \tau\}$ the current favorable region. To keep transport path inside $\mathcal{F}_0$, we use the intrinsic distance

$$d_{\mathcal{F}_0}(z,y) = \inf_{\substack{\gamma(0)=z,\ \gamma(1)=y \\ \gamma(t)\in\mathcal{F}_0,\ \forall t\in[0,1]}} \int_0^1 \|\dot{\gamma}(t)\|_2 \, dt.$$

and solve

$$\pi^\star \in \arg\min_{\pi \in \Pi(\mathbb{P}_{\mathrm{CF}}, \mathbb{P}_1)} \sum_{i,j} \pi_{ij} d^2_{\mathcal{F}_0}(z_i, y_j), \tag{9}$$

where $\Pi(\mathbb{P}_{\mathrm{CF}}, \mathbb{P}_1)$ denotes the set of couplings with marginals $\mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1$. By construction $d_{\mathcal{F}_0} \ge \|\cdot\|_2$, with equality on $\mathcal{F}_0$ exactly when $\mathcal{F}_0$ is convex, i.e., the straight-line segment between any $z, y \in \mathcal{F}_0$ is feasible, in which case $d_{\mathcal{F}_0}(z,y) = \|z - y\|_2$. The refinement is the displacement interpolation along the minimizing intrinsic transport paths: $\mathbb{P}_{\mathrm{rCF}} = \sum_{i=1}^n \sum_{j=1}^m \pi_{ij}^\star \, \delta_{\gamma_{ij}^\star(\lambda^\star)}$, where $\gamma_{ij}^\star : [0,1] \to F_0$ is a constant-speed d_{F_0}-minimizing geodesic from z_i to y_j. When F_0 is convex, $\gamma_{ij}^\star(\lambda) = (1-\lambda)z_i + \lambda y_j$, recovering standard Euclidean displacement interpolation. By the equivalence between McCann displacement interpolation and the two-measure Wasserstein barycenter (Agueh & Carlier, 2011), $\mathbb{P}_{\mathrm{rCF}} = \mathbb{Q}(\lambda^\star)$, where $\mathbb{Q}(\lambda)$ is the certified family derived in Proposition 3.

This realization separates the two roles in LIFT-CF: $\pi^\star$ determines how probability mass is transported between the original counterfactual and target distributions, whereas $\lambda^\star$ determines how far that mass moves along the resulting displacement paths, thereby also determining the consistency–plausibility trade-off. Thus, once $\lambda^\star$ has been selected by deployment calibration, no additional refinement parameter is required.

Table 1: Robustness comparison under matched recourse budgets. Color indicates improvement; darker green implies greater improvement.

Dataset	Baseline	Matched L2 ↓	Baseline Mean Future ↑	LIFT-CF Mean Future ↑
HELOC	T-Rex	1.961	74.32%	77.15%
	ROAR	2.134	71.05%	74.22%
	PROPLACE	2.456	69.87%	73.01%
	ElliCE	1.823	73.44%	76.33%
GERMAN CREDIT	T-Rex	1.542	68.50%	71.20%
	ROAR	1.876	65.32%	68.47%
	PROPLACE	2.103	62.18%	65.89%
	ElliCE	1.654	66.77%	69.55%
FICO	T-Rex	2.110	72.90%	75.40%
	ROAR	2.345	70.12%	73.28%
	PROPLACE	2.678	68.44%	71.56%
	ElliCE	1.987	71.33%	74.05%
DIABETES	T-Rex	1.234	75.60%	78.20%
	ROAR	1.456	72.85%	75.90%
	PROPLACE	1.678	70.22%	73.45%
	ElliCE	1.345	74.10%	76.85%

4 EXPERIMENTS

We evaluate LIFT-CF as a post-hoc distributional robustness wrapper for counterfactual explanation generators, across four real-world datasets.

Datasets. We evaluate LIFT-CF on three widely used tabular classification benchmarks: HELOC, GERMAN CREDIT, FICO, and DIABETES.

Baselines. We compare LIFT-CF against four representative methods for robust counterfactual generation: T-Rex, ROAR, PROPLACE, and ElliCE. T-Rex targets robustness to naturally occurring model changes, while ROAR formulates robust recourse through an adversarial minimax objective over model shifts. PROPLACE uses robust optimization to generate counterfactuals that are both provably robust to bounded model changes and plausible with respect to the data distribution, whereas ElliCE constructs robust recourse over an ellipsoidal approximation of the Rashomon set of near-optimal models. These methods provide complementary baselines spanning empirical, adversarial, and certified notions of robustness. In contrast, LIFT-CF does not replace the underlying counterfactual generator or explicitly optimize against a prescribed model-parameter uncertainty set. Instead, it operates as a post-hoc wrapper that refines counterfactuals produced by a base generator at the distribution level. All methods are evaluated using the same factual instances and held-out future-model shifts, with robustness reported jointly with recourse cost.

Future model shifts. We evaluate robustness under future model updates induced by shifts in the training distribution. For each dataset, we independently generate mixed perturbations of the original training distribution and retrain the predictive model either from scratch or by fine-tuning on each shifted dataset, yielding a collection of future classifiers. These evaluation shifts are constructed independently of the bootstrap procedure used for calibration: the bootstrap determines only the reference stress scale b_{cal} and does not specify the direction or mechanism of the future shift. We parameterize evaluation severity by the relative shift magnitude r/b_{cal}. Our primary comparison uses matched-scale shifts with $r/b_{\text{cal}} = 1$, while additional stress tests vary r/b_{cal} beyond the calibration scale to assess robustness degradation under increasingly severe shifts. Counterfactuals are generated before any evaluation shift is observed and remain fixed thereafter. All methods are evaluated on exactly the same shifted datasets and retrained future classifiers, providing a common external evaluation protocol despite their different construction-time uncertainty models.

Evaluation metrics. We evaluate counterfactual quality along three primary dimensions: validity, robustness, and recourse cost. *Validity* measures the fraction of generated counterfactuals that achieve the desired prediction under the original classifier. *Future validity* measures robustness to model updates and is defined as the fraction of held-out future classifiers under which a counterfactual retains the desired prediction; we average this quantity across counterfactual instances and

future models. *Recourse cost* is measured by the average ℓ_2 distance between each factual instance x and its counterfactual x_{cf}, i.e., $\frac{1}{N}\sum_{i=1}^{N}\|x_i - x_{\mathrm{cf},i}\|_2$. Since robustness can often be increased by moving counterfactuals farther from their factual instances, we report future validity jointly with recourse cost and interpret robustness improvements relative to the additional cost required to obtain them.

Experimental setup. For each dataset, we use a stratified cross-validation protocol that separates model fitting, operating-point calibration, radius selection, and final robustness evaluation. Within each fold, we first train a base classifier f_{base}. CFs are generated for instances predicted as belonging to the unfavorable class by f_{base}. For LIFT-CF, the resulting base CFs define the empirical proximal anchor P_{CF}, while favorable training samples define the empirical target anchor P_1. These quantities determine the Wasserstein barycentric refinement path $\{\mathbb{Q}(\lambda) : \lambda \in [0,1]\}$. We consider a predefined set of Wasserstein uncertainty radii $\mathcal{R} = \{r_1, \ldots, r_m\}$. For each fixed $r \in \mathcal{R}$, we use the training-derived Wasserstein adversarial search described in Section **??** to construct a radius-specific adverse model f_r^{adv}. All adverse models are obtained by fine-tuning from the same fitted classifier f_{base} and are frozen before the calibration split is accessed. We then sweep λ and evaluate, on the calibration split, both the recourse cost and the validity of the refined CFs under these radius-specific proxy models. This yields, for each r, a robustness–proximity frontier $\left\{\left(C_{\mathrm{cal}}(\lambda), V_{\mathrm{cal}}(r,\lambda)\right) : \lambda \in \Lambda\right\}$, where lower recourse cost and higher calibration validity are preferred. A predefined frontier operating-point rule is used to select one refinement depth $\lambda_r^\star$ for each candidate radius. The resulting configurations $\{(r, \lambda_r^\star) : r \in \mathcal{R}\}$ are then compared on a disjoint shift-validation set that is not used for the within-radius selection of $\lambda_r^\star$. This second-stage validation is used to select the final uncertainty radius $r^\star$, after which the corresponding refinement depth $\lambda_{r^\star}^\star$ is fixed. No information from the final future-shift test set is used in either λ- or r-selection. Final robustness is evaluated on a separate collection of held-out future distributions. Each future classifier is obtained by fine-tuning f_{base} on its corresponding shifted training distribution using a fixed update protocol. LIFT-CF and all baselines are evaluated on the same factual instances and the same future-model realizations. We report future validity, recourse proximity, and favorable-population alignment. For matched-cost baseline comparisons, we additionally evaluate LIFT-CF under the recourse cost attained by the calibration-selected operating point of each baseline; this matched-cost protocol is used only for head-to-head evaluation and is separate from the LIFT-CF deployment-selection procedure.

5 Conclusion

We introduced LIFT-CF, a post-hoc distributional robustness wrapper for counterfactual explanation generators. Rather than certifying individual counterfactuals against future model changes, LIFT-CF refines the population of counterfactuals produced by a base generator along a certified Wasserstein barycentric path toward the observed target-class distribution.

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A Appendix

A.1 Robust Risk Bounds and the Certified Barycentric Family

This appendix provides the complete derivation underlying Proposition 3. The argument adapts the CF-consistent Wasserstein robust-bound construction to the present post-hoc refinement setting.

Proof of Proposition 3. Recall the uncertainty lens

$$\mathcal{C} = \{\mu : W_2(\mu, \mathbb{P}_1) \le \epsilon,\ W_2(\mu, \mathbb{P}_{\mathrm{CF}}) \le \delta\}, \tag{10}$$

and the corresponding worst-case risk

$$R(\mathbb{Q}) = \sup_{\mu \in \mathcal{C}} W_2^2(\mathbb{Q}, \mu). \tag{11}$$

Fix any $\mu \in \mathcal{C}$. By the triangle inequality,

$$\begin{aligned} W_2(\mathbb{Q}, \mu) &\le W_2(\mathbb{Q}, \mathbb{P}_{\mathrm{CF}}) + W_2(\mathbb{P}_{\mathrm{CF}}, \mu) \\ &\le W_2(\mathbb{Q}, \mathbb{P}_{\mathrm{CF}}) + \delta, \end{aligned} \tag{12}$$

and similarly,

$$\begin{aligned} W_2(\mathbb{Q}, \mu) &\le W_2(\mathbb{Q}, \mathbb{P}_1) + W_2(\mathbb{P}_1, \mu) \\ &\le W_2(\mathbb{Q}, \mathbb{P}_1) + \epsilon. \end{aligned} \tag{13}$$

Therefore,

$$W_2^2(\mathbb{Q}, \mu) \le \min\left\{ \left(W_2(\mathbb{Q}, \mathbb{P}_{\mathrm{CF}}) + \delta\right)^2,\ \left(W_2(\mathbb{Q}, \mathbb{P}_1) + \epsilon\right)^2 \right\}. \tag{14}$$

For any $\alpha \in [0,1]$ and any $a, b \geq 0$,

$$\min\{a,b\} \leq (1-\alpha)a + \alpha b. \tag{15}$$

Applying this inequality to Eq. 14 gives

$$\begin{aligned} W_2^2(\mathbb{Q},\mu) \leq (1-\alpha)\left(W_2(\mathbb{Q},\mathbb{P}_{\mathrm{CF}}) + \delta\right)^2 \\ + \alpha\left(W_2(\mathbb{Q},\mathbb{P}_1) + \epsilon\right)^2. \end{aligned} \tag{16}$$

We next control the cross terms. For any $\gamma > 0$ and $a, b \geq 0$, Young's inequality gives

$$2ab \leq \gamma a^2 + \frac{b^2}{\gamma}. \tag{17}$$

Hence, for arbitrary $\gamma_0, \gamma_1 > 0$,

$$\begin{aligned} \left(W_2(\mathbb{Q},\mathbb{P}_{\mathrm{CF}}) + \delta\right)^2 \leq (1+\gamma_0) W_2^2(\mathbb{Q},\mathbb{P}_{\mathrm{CF}}) \\ + \left(1 + \frac{1}{\gamma_0}\right)\delta^2, \end{aligned} \tag{18}$$

and

$$\begin{aligned} \left(W_2(\mathbb{Q},\mathbb{P}_1) + \epsilon\right)^2 \leq (1+\gamma_1) W_2^2(\mathbb{Q},\mathbb{P}_1) \\ + \left(1 + \frac{1}{\gamma_1}\right)\epsilon^2. \end{aligned} \tag{19}$$

Combining Eqs. 16–19, we obtain, for every $\mu \in \mathcal{C}$,

$$\begin{aligned} W_2^2(\mathbb{Q},\mu) \leq (1-\alpha)(1+\gamma_0) W_2^2(\mathbb{Q},\mathbb{P}_{\mathrm{CF}}) \\ + \alpha(1+\gamma_1) W_2^2(\mathbb{Q},\mathbb{P}_1) + K(\alpha,\gamma_0,\gamma_1), \end{aligned} \tag{20}$$

where

$$\begin{aligned} K(\alpha,\gamma_0,\gamma_1) := (1-\alpha)\left(1 + \frac{1}{\gamma_0}\right)\delta^2 \\ + \alpha\left(1 + \frac{1}{\gamma_1}\right)\epsilon^2. \end{aligned} \tag{21}$$

Importantly, $K(\alpha,\gamma_0,\gamma_1)$ is independent of $\mathbb{Q}$.

Since Eq. 20 holds for every $\mu \in \mathcal{C}$, taking the supremum over the lens yields

$$\begin{aligned} R(\mathbb{Q}) \leq (1-\alpha)(1+\gamma_0) W_2^2(\mathbb{Q},\mathbb{P}_{\mathrm{CF}}) \\ + \alpha(1+\gamma_1) W_2^2(\mathbb{Q},\mathbb{P}_1) + K(\alpha,\gamma_0,\gamma_1). \end{aligned} \tag{22}$$

Define

$$A := (1-\alpha)(1+\gamma_0), \qquad B := \alpha(1+\gamma_1). \tag{23}$$

For $A + B > 0$, the $\mathbb{Q}$-dependent part of Eq. 22 can be written as

$$\begin{aligned} A\, W_2^2(\mathbb{Q},\mathbb{P}_{\mathrm{CF}}) + B\, W_2^2(\mathbb{Q},\mathbb{P}_1) \\ = (A+B)\left[(1-\lambda) W_2^2(\mathbb{Q},\mathbb{P}_{\mathrm{CF}}) + \lambda W_2^2(\mathbb{Q},\mathbb{P}_1)\right], \end{aligned} \tag{24}$$

where

$$\lambda := \frac{B}{A+B} = \frac{\alpha(1+\gamma_1)}{(1-\alpha)(1+\gamma_0) + \alpha(1+\gamma_1)}. \tag{25}$$

Because the additive term K is independent of $\mathbb{Q}$ and the multiplicative factor $A + B$ is positive, minimizing the robust upper bound with respect to $\mathbb{Q}$ is equivalent to minimizing

$$(1-\lambda) W_2^2(\mathbb{Q},\mathbb{P}_{\mathrm{CF}}) + \lambda W_2^2(\mathbb{Q},\mathbb{P}_1). \tag{26}$$

Thus, for every admissible choice of the bounding parameters, the corresponding robust upper bound is minimized by a 2-Wasserstein barycenter of $\mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1$ with weights $(1-\lambda, \lambda)$.

It remains to show that the induced barycentric weight spans the entire interval $[0, 1]$. Fix arbitrary $\gamma_0, \gamma_1 > 0$ and define

$$\lambda(\alpha) = \frac{\alpha(1+\gamma_1)}{(1-\alpha)(1+\gamma_0) + \alpha(1+\gamma_1)}. \tag{27}$$

This mapping is continuous on $[0, 1]$, with

$$\lambda(0) = 0, \qquad \lambda(1) = 1. \tag{28}$$

Moreover,

$$\frac{d\lambda(\alpha)}{d\alpha} = \frac{(1+\gamma_0)(1+\gamma_1)}{\left[(1-\alpha)(1+\gamma_0) + \alpha(1+\gamma_1)\right]^2} > 0. \tag{29}$$

Hence $\lambda(\alpha)$ is strictly increasing and maps $[0, 1]$ onto $[0, 1]$.

Consequently, the minimizers of this family of valid robust upper bounds span the Wasserstein barycentric family

$$\mathcal{Q}_{\mathrm{bar}} = \{\mathbb{Q}(\lambda) : \lambda \in [0, 1]\}, \tag{30}$$

which proves Proposition 3. □

A.2 Proof of the Monotone Transport Geometry

Proof of Proposition **??**. Let

$$t = \eta\lambda^\star.$$

By Eq. **??**, $\mathbb{P}^{(\eta)}_{\mathrm{rCF}}$ is the displacement interpolation at time t induced by an optimal coupling between $\mathbb{P}_{\mathrm{CF}}$ and $\mathbb{P}_1$.

Since displacement interpolation under the quadratic Wasserstein metric defines a constant-speed geodesic, for any $t \in [0, 1]$,

$$W_2(\mathbb{P}_{\mathrm{CF}}, \mathbb{Q}(t)) = t\, W_2(\mathbb{P}_{\mathrm{CF}}, \mathbb{P}_1), \tag{31}$$

$$W_2(\mathbb{Q}(t), \mathbb{P}_1) = (1-t)\, W_2(\mathbb{P}_{\mathrm{CF}}, \mathbb{P}_1). \tag{32}$$

Substituting $t = \eta\lambda^\star$ and $D = W_2(\mathbb{P}_{\mathrm{CF}}, \mathbb{P}_1)$ gives

$$W_2\left(\mathbb{P}^{(\eta)}_{\mathrm{rCF}}, \mathbb{P}_{\mathrm{CF}}\right) = \eta\lambda^\star D, \tag{33}$$

$$W_2\left(\mathbb{P}^{(\eta)}_{\mathrm{rCF}}, \mathbb{P}_1\right) = (1-\eta\lambda^\star)D. \tag{34}$$

The first quantity is nondecreasing in η, while the second is nonincreasing in η. Summing the two identities yields Eq. **??**. □

A.3 Proof of Proposition **??**

Proof. Since f is L_f-Lipschitz, the surrogate $\ell_f(z) = [\tau - f(z)]_+$ is also L_f-Lipschitz. Hence, by the Kantorovich–Rubinstein inequality,

$$\mathbb{E}_{z\sim\mathbb{Q}}[\ell_f(z)] \le \mathbb{E}_{z\sim\mu_f}[\ell_f(z)] + L_f W_1(\mathbb{Q}, \mu_f). \tag{35}$$

Using $W_1 \le W_2$ and Assumption **??**,

$$\mathbb{E}_{z\sim\mathbb{Q}}[\ell_f(z)] \le \alpha_f + L_f W_2(\mathbb{Q}, \mu_f). \tag{36}$$

Since $\mu_f \in \mathcal{C}$ and

$$R(\mathbb{Q}) = \sup_{\mu\in\mathcal{C}} W_2^2(\mathbb{Q}, \mu),$$

we have

$$W_2(\mathbb{Q}, \mu_f) \le \sqrt{R(\mathbb{Q})}.$$

Therefore,

$$\mathbb{E}_{z\sim\mathbb{Q}}[\ell_f(z)] \le \alpha_f + L_f\sqrt{R(\mathbb{Q})}, \tag{37}$$

which proves the result. □